Seat No. _____ Enrollment No. _____

THE CHARUTAR VIDYA MANDAL UNIVERSITY

B.E.– SEMESTER 6

APRIL 2023 (REGULAR) EXAMINATION

Course Title: Smart Cities Planning And Management

Course Code: 102030622

Total Printed Pages: 02

Date: 05/05/2023 **Time: 10.00 am to 12.30 pm** **Maximum Marks: 60**

Instructions:

- Attempt all questions.
- Numbers to the right indicate full marks for each question.
- Make suitable assumption if required, do specify the same.

Q: 1 Answer the Following. (Each question carries one marks). **[10]**

(i) Choose the correct option for Smart City.
1. A 'smart city' is an urban region that is highly advanced in terms of overall infrastructure, sustainable real estate, communications and market viability.
2. It will provide real time information on parking, traffic congestion, public transport.
3. Smart cities will be energy efficient and will have low carbon foot print.

a) 1,2 b) 2,3 c) 1,3 d) All

(ii) Smart Cities Mission comprising 100 Indian Cities was launched in _____.
a) 2015 b) 2014 c) 2017 d) 2020

(iii) ITS stands for _____.
a) Internet Travel Services b) Internet Transportation Security
c) Intelligent Transportation Security d) Intelligent Transport System

(iv) Write the full form of **any one** : SCADA/AMRUT

(v) Smart Recreation is a part of Smart City Programme Model?
a) True b) False

(vi) What is the example of Smart grid edge device for utility?
a) Smart Meters b) Smart Homes c) Smart Car d) Smart Collage

(vii) The main purpose of Rainwater harvesting is recharge the _____.
a) Spring b) Well c) Canal d) Ground water

(viii) Which of the following is not the purpose of Green Buildings?
a) To reduce use of water b) To minimize damage to the environment
c) Reuse of waste materials d) None of the above

(ix) LEED and GRIHA are the Green Rating Systems currently working in India.
a) True b) False

(x) Which one is not a method for water conservation?
a) Drip Irrigation system b) Groundwater extraction c) Kunds d) Avoiding water wastage

Q: 2 **(A)** Discuss the different types of Infrastructure Systems (Pillars) of a Smart City. **[04]**

(B) Explain the features of Smart Cities. **[06]**

Q: 3 **(A)** Discuss case study of **ANYONE** smart city: (1) GIFT city (2) Ahmedabad (3) DHOLERA SIR (4) Surat **[05]**

(B) Explain Housing for Smart Cities. Discuss the criteria for it. **[05]**

OR

(B) What are the purposes and applications of ITS? **[05]**

Q: 4 **(A)** Explain the features of Green Buildings. **[05]**

(B) What is Smart Grid? What is the need of it in India? Write characteristics of Smart Grid. **[05]**

OR

(B) Discuss the key technologies needed for the Smart City development. **[05]**

Q: 5 **(A)** Discuss various phases of Smart City project management. **[05]**

(B) Describe various applications of GIS and GPS in Smart Cities. **[05]**

OR

(B) Write a note on Flood Management in Smart Cities. **[05]**

Q: 6 **(A)** Enlist and explain various methods of water resource conservation. **[05]**

(B) What are the problems present in sanitation practices? Discuss the innovative sanitation solutions for Smart Cities. **[05]**

OR

(B) Write a note on : 1) e-Ticketing 2) Smart mobility **[05]**

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